

Auteur: Stan Van Campenhout

Studierichting: Informatica

Oefeningenreeks 1C nr. 44.8

$$z^2 = 1 + \sqrt{3}i$$

$$\text{modulus: } r = \sqrt{1^2 + \sqrt{3}^2} = \sqrt{4} = 2$$

$$\text{argument: } \tan \theta = \frac{\sqrt{3}}{1} \Rightarrow \theta = \arctan \sqrt{3} = \frac{\pi}{3}$$

$$z^2 = 2 \operatorname{cis} \left(\frac{\pi}{3} + k \cdot 2\pi \right) \Rightarrow z = \sqrt{2} \operatorname{cis} \left(\frac{\frac{\pi}{3} + k \cdot 2\pi}{2} \right) \quad \forall k \in \{0, 1\}$$

$$k = 0 \rightarrow \sqrt{2} \operatorname{cis} \left(\frac{\frac{\pi}{3} + 0 \cdot 2\pi}{2} \right) = \sqrt{2} \operatorname{cis} \left(\frac{\pi}{6} \right) = \frac{\sqrt{6}}{2} + \frac{\sqrt{2}}{2}i$$

$$k = 1 \rightarrow \sqrt{2} \operatorname{cis} \left(\frac{\frac{\pi}{3} + 1 \cdot 2\pi}{2} \right) = \sqrt{2} \operatorname{cis} \left(\frac{7\pi}{6} \right) = -\frac{\sqrt{6}}{2} - \frac{\sqrt{2}}{2}i$$

References